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A low-cost spectrometer for NMR measurements in the Earth's magnetic field

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Abstract

We describe and demonstrate an inexpensive, easy-to-build, portable spectrometer for nuclear magnetic resonance measurements in the Earth's magnetic field. The spectrometer is based upon a widely available inexpensive microcontroller, which acts as a pulse programmer, audio-frequency synthesizer and digitizer, replacing what are typically the most expensive specialized components of the system. The microcontroller provides the capability to execute arbitrarily long and complicated sequences of phase-coherent, phase-modulated excitation pulses and acquire data sets of unlimited duration. Suitably packaged, the spectrometer is amenable to measurements in the research lab, in the field or in the teaching lab. The choice of components was heavily weighted by cost and availability, but required no significant sacrifice in performance. Using an existing personal computer, the resulting design can be assembled for as little as US\$200. The spectrometer performance is demonstrated with spin-echo and Carr–Purcell–Meiboom–Gill pulse sequences on a water sample.

Keywords: mobile NMR, arduino microcontroller, magnetic resonance imaging, magnetometry

1. Introduction

Nuclear magnetic resonance (NMR) is a widely used technique for the study of materials of all kinds. Most modern NMR spectrometers are based on superconducting magnets, and operate with magnetic fields up to ~ 20 T at frequencies up to 1 GHz. At the other end of the frequency regime, however, NMR in the Earth's field requires no superconducting magnet, but generally delivers much less information about the sample due to the loss of chemical shift resolution. Low-frequency NMR, however, continues to be a subject of active research. Some current applications include studies of the liquid brine phase of sea ice in Antarctica [1–3] motivated by the importance of sea ice in global climate, the measurement of heteronuclear J-couplings [4, 5], the recent demonstrations that homonuclear J-couplings and chemical shifts can be observed at low field [6, 7], and the study of nuclear spin relaxation times in tissues, as at low fields T_1 times appear to be a promising

way of identifying tumors [8]. Research on NMR in the Earth's field has been recently reviewed [9].

Instrumentation for NMR at various magnetic field strengths, including the Earth's field, is available commercially, but commercial instruments are generally rather expensive due to the use of highly specialized components. Most commercial NMR systems incorporate a proprietary pulse programmer and discrete frequency synthesizers. In this work we describe a simple Earth's field NMR spectrometer, in which the expensive specialized components are replaced with inexpensive, easy-to-find, general-purpose, mass-market components. While the system constructed here is intended for use in a teaching lab, it could be easily adapted for use in the research lab, or in the field. Two of the most important aspects of this device are its low cost ($\sim$ US\$200, plus a notebook computer) and simplicity. Despite the extraordinarily low cost, few sacrifices, and in fact some gains, are made in the performance and flexibility of the device compared to similar research apparatus previously reported. This implementation

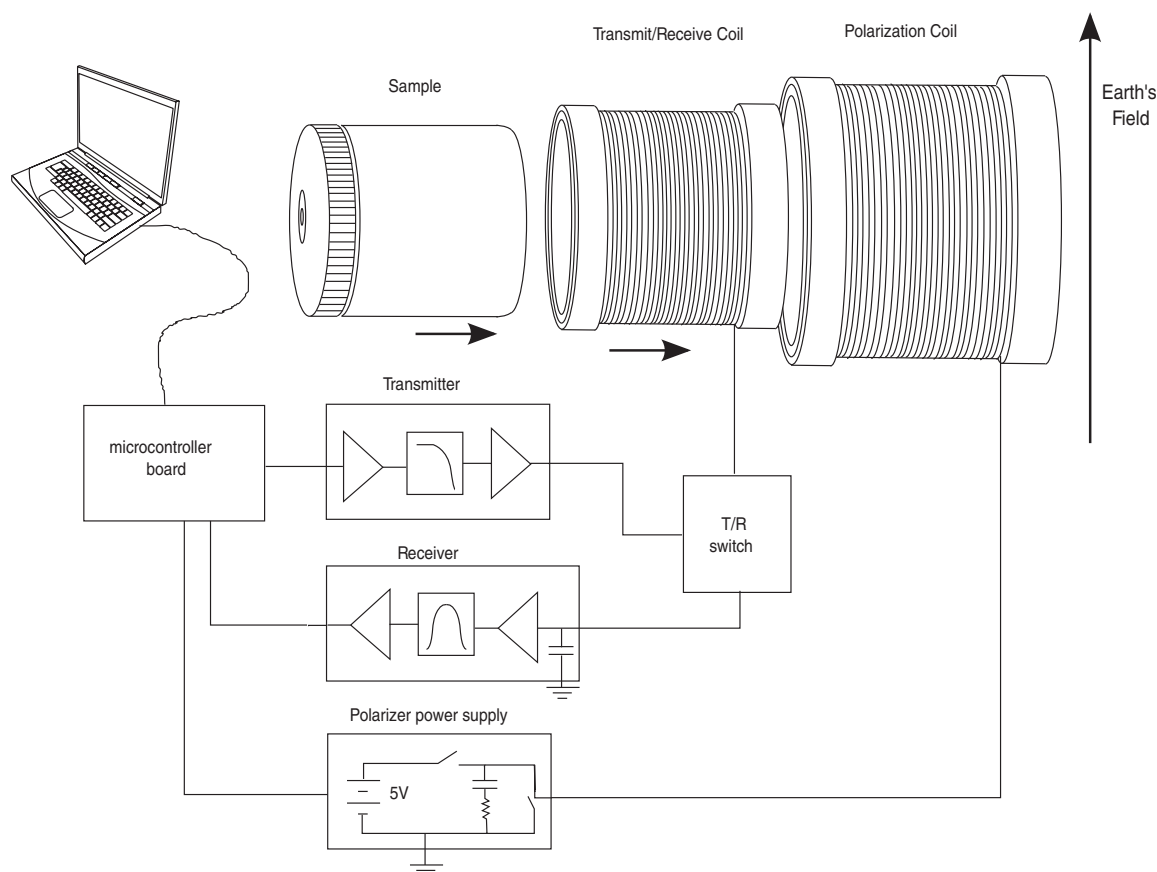


Figure 1. Block diagram of the Earth's field NMR spectrometer

makes NMR spectroscopy available to any researcher or student.

2. Description

A simple pulsed-NMR experiment begins with the application of a near-resonant excitation pulse applied to a coil containing the sample of interest. This pulse tips the nuclear magnetization from its equilibrium orientation, along a static magnetic field, into the plane perpendicular to the field. Following the pulse, precession of the magnetization about the static field induces a voltage in a receive coil. This signal is amplified, filtered and digitized for further processing and display. One attractive feature of NMR in the Earth's field is that if care is taken in siting and choice of materials, the field over the sample volume can be very uniform, giving NMR linewidths significantly less than 1 Hz. Because the net nuclear magnetization in thermal equilibrium at room temperature in low field is very small, such NMR experiments are often preceded by a polarization period, during which a much larger, but potentially more inhomogeneous, magnetic field is applied in order to increase the net polarization of the nuclear spins.

More sophisticated NMR experiments may involve applying more pulses, and demand that the phase and/or amplitude of the pulses be modulated. Signal averaging, which is often required to increase the signal-to-noise ratio

(SNR) of the received signals, requires a stable frequency source and reproducible pulse and analog-to-digital converter (ADC) timings.

The device to be described is a simple but versatile NMR spectrometer capable of executing arbitrarily long and complex pulse sequences reproducibly with sub-microsecond timing resolution. It can produce trains of phase-coherent audio-frequency pulses that can be phase modulated, and can digitize signals with 10 bit resolution at nearly 10 kHz for arbitrarily long periods.

The design of the present device draws deeply on that of Callaghan *et al* [10], but simplifies some aspects of their design, and replaces the most expensive specialized components with inexpensive, flexible and widely available ones. Most significantly, Callaghan *et al* used a specialized NMR console as a pulse programmer, audio-frequency synthesizer and digitizer. Here, we replace this expensive component with a small, easily acquired \$30 microcontroller board. Other components of the present design are also optimized for ease of acquisition and cost. The present device can easily be built by anyone competent with a soldering iron.

A block diagram of the spectrometer is shown in figure 1. The present design uses two coils, one transmit/receive coil used to apply audio-frequency pulses and detect the NMR signals, and a second, larger coil for polarizing the nuclear spins. Details of the coils are collected in table 1. The transmit/receive coil is perhaps the most challenging component of the spectrometer to construct, as it consists

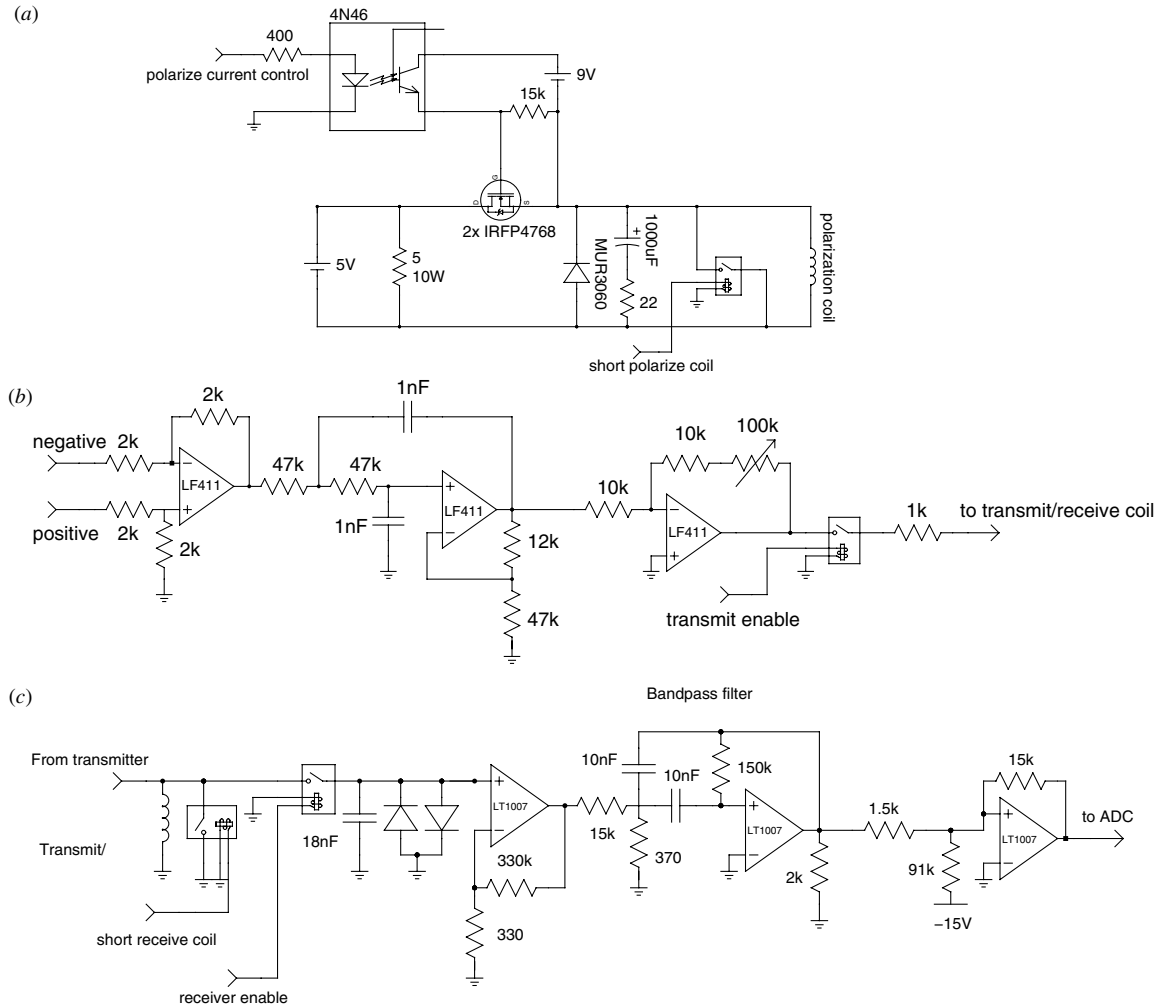


Figure 2. Schematic diagrams showing the circuits used for (a) the polarizing coil driver, (b) the audio-frequency transmitter and (c) the receiver. All relays are Hamlin HE3351A055.

Table 1. Coil details.

	T/R coil	Polarization coil
Diameter (cm)	10	12
Length (cm)	10	15
Wire size (AWG)	30	18
Layers	10	3 (in parallel)
Turns per layer	310	125
Inductance	700 mH	900 μ H
Resistance (Ω)	310	0.34

of 10 layers and some 3100 turns of AWG 30 copper wire (McMaster-Carr, Los Angeles, CA), wound on a section of 3" ABS drain-waste-vent (DWV) pipe, acquired from a local building supply store. This coil was wound on a Meteor ME 301 coil winding machine (Meteor AG, Wollerau, Switzerland). The sample is placed in an 0.55 L polycarbonate bottle (Nalgene, Rochester, NY) which fits snugly inside the ABS form.

The polarization coil consists of three layers of AWG 18 copper wire, wound by hand on a section of 4" ABS DWV pipe. The windings of the polarization coil are connected in parallel to allow the use of a 5 V power supply

that had been removed from an old personal computer. In operation, the coil draws 13.4 A and produces a field of approximately 10.5 mT, about 200 times the Earth's field. The current to this coil is switched by means of a pair of high-current MOSFETs (IRFP4768, International Rectifier, El Segundo, CA). The nuclear magnetization, which develops along the coil axis during polarization, is adiabatically rotated to point along the Earth's field by turning off the current over ~ 10 ms. Another alternative would be to turn off the polarization current suddenly, leaving the magnetization pointing perpendicular to the Earth's field. A rapid enough turn-off of the polarization coil is easily achieved (by charging a high-voltage capacitor through a diode when the MOSFETs turn off) if the polarization coil is isolated; however, with the much higher inductance of the detection coil strongly coupled to the polarization coil, producing such a rapid turn-off is problematic, as the detection coil is self-resonant at about 5 kHz. A schematic of the circuit used to drive and shut off the polarization coil is shown in figure 2(a). After the MOSFETs turn off, the current in the coil is turned off exponentially with a time constant of RC . This resistor/capacitor combination, with the polarization coil inductance, forms an over-damped series RLC circuit.

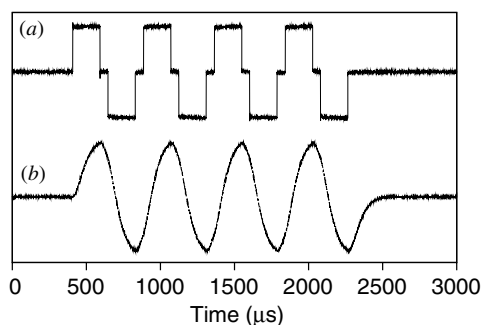


Figure 3. Transmitter waveforms. (a) The output from the summing amplifier, and (b) the result following the low-pass filter.

Use of an opto-isolator to drive the MOSFETs allows one side of the coil to be grounded at all times, and the 9 V supply used to power the MOSFETs is simply a 9 V alkaline battery. The relay allows the coil to be shorted, which, as suggested by Callaghan *et al* [10], helps to shield the detection coil from audio-frequency pickup. The 5 Ω , 10 W resistor is required by the power supply. Most personal computer power supplies have maximum and minimum loads beyond which they shut down. This 5 Ω resistor draws 1 A, enough to keep the supply used here in its operating range when the coil is not drawing current.

A diagram of the audio-frequency transmitter is shown in figure 2(b). Two logic lines from the microcontroller board are used to generate positive and negative lobes of the transmitter waveform to allow accurate phase cycling of the audio-frequency signal. These two signals are combined by an op-amp subtraction circuit, passed through a low-pass Bessel filter, and amplified or attenuated to the desired level. Waveforms before and after the filter, acquired with a Tektronix 3052 digital oscilloscope, are shown in figure 3. The duty cycle of the digital drive is restricted in order to reduce the harmonic content of the sum signal.

The receiver, figure 2(c), consists of three low-noise op-amps (LT1007, Linear Technology, Milpitas, CA), one of which is configured as a band-pass filter. The final op-amp adds a 2.5 V dc offset to shift to the middle of the 0–5 V ADC voltage range. The LT1007 was chosen for its noise performance, low cost, and the fact that it uses the same pin configuration as many standard op-amps (e.g. 741 and 411). This last aspect is expected to be important in a teaching lab, where a common pin-out should help to prevent op-amp mis-wiring. The overall gain of the receiver is about 50 000. The receive coil is tuned to resonance with a fixed 18 nF capacitor located at the receiver.

The transmit/receive switch consists of three small reed relays, whose drive requirements (40 mA) are satisfied by the digital outputs of the microcontroller. The digital outputs driving the relays are protected from voltage spikes on relay shut-off by small signal diodes (1N4148, not shown) connected across each relay coil. Upon switching between transmit and receive, a receiver dead-time of 10 ms is found between the end of the pulse and the beginning of signal acquisition. The order of magnitude of this time is in agreement with that expected, as a resonant circuit having a Q of 5.8 (see the appendix) at

a frequency of 2 kHz would have a decay time constant of 2.8 ms. Thus, any switching transients can be expected to decay on this time scale.

The microcontroller board, which acts here as a pulse programmer, audio-frequency synthesizer and digitizer, is an Arduino Duemilanove, based on the ATmega328 microcontroller (Atmel, San Jose, CA). Arduino is an open-source hardware and software platform that greatly simplifies the use of the ATmega. Arduino boards are available from numerous vendors, and a complete integrated development environment is freely available [11]. Arduino boards incorporate a USB to serial converter chip, allowing a single USB connection to the host computer, through which programs are burned into the microcontroller flash, communications with the board are established when operating and power to the board may be supplied. With the 16 MHz crystal oscillator on the Duemilanove, the ADC achieves its maximal 10 bit resolution at a maximum sample rate of 9615 samples s^{-1} , more than adequate for capturing the ~ 2 kHz signals in the Earth's field. Because the ADC sampling frequency is more than twice the signal frequency and all the signals appear within a bandwidth much smaller than the ADC sample rate, no quadrature detection is necessary. All of the information that would be available from a quadrature receiver is retained by the single-channel ADC.

The microcontroller board, if programmed carefully, is capable of producing trains of phase coherent pulses with reproducible timing, allowing not only accurate signal averaging, but phase cycling as well. The frequency stability of the system is determined by the stability of the oscillator driving the microcontroller. Here, the oscillator is an inexpensive crystal oscillator, Citizen HC49US (supplied with the Arduino board). This oscillator has a specified stability of ± 50 ppm over the temperature range of -10 to 60 $^{\circ}C$, corresponding to a variation in the measured resonance frequency of about 0.2 Hz over this temperature range. Because the frequency of such crystal oscillators depends on temperature in a deterministic way, the stability will be much better than this for more modest temperature variations. For field-work under extreme conditions, a temperature controlled oscillator could be employed.

For the experiments demonstrating the device below, the pulse-programming logic is performed in real time on the microcontroller. While the programs written to date have used only a very small fraction of the available memory of the microcontroller, it is conceivable that a complicated pulse program would either not fit in the microcontroller's available memory, or would be too computationally intensive to run in real time. Either situation however could be avoided by employing the microcontroller essentially as a state machine, and downloading pre-calculated states and durations from the host computer. This would allow arbitrarily long and complex pulse sequences to be executed. Data from the microcontroller's ADC is uploaded 'on the fly' to the host computer allowing as many data points to be acquired as desired.

Programs that execute on the Arduino are written in C/C++, and a simple C program was written to execute on

Table 2. Major components.

Component	Cost	Source
Arduino Duemilanove	\$30	Sparkfun Electronics
Aluminum shielding	\$42	Metal Supermarkets
18 # copper wire	\$28	McMaster-Carr
30 # copper wire	\$23	McMaster-Carr
LT1007 op-amps	3 × \$3	Digikey Corporation
LF411 op-amps	3 × \$2	Newark Electronics
HE3351A055 relays	4 × \$3	Newark Electronics
IRF4768 MOSFET	2 × \$9	Newark Electronics
MUR3060 diode	\$5	Digikey Corporation
4N46 optoisolator	\$4	Digikey Corporation
ABS pipe and fittings	\$10	Local building supply store
PC power supply	\$0–\$15	used (or e.g. newegg.com)

the host computer to trigger sequence start and receive the acquired data. Both the microcontroller code and the program used to communicate with it have been made available at <http://www.phas.ubc.ca/~michal/Earthsfield>.

A list of all of the major components in the spectrometer is presented in table 2. This list sums to \$187, and does not include resistors, capacitors, small-signal diodes, hook-up wire, or breadboards on which to build the circuits. In addition, a computer is required to program the microcontroller and accept the digital data. The Arduino software is multiplatform, available for Windows, Mac OS X and Linux. The computer required here must run a supported operating system and have an available USB port.

The problem of audio-frequency pickup is ubiquitous in low-frequency NMR, and has generally been handled either by extensive shielding or by the use of two opposing coils, so that the pickup of environmental noise from the two coils cancels [9]. We found the shielding provided by shorting the polarization coil to be insufficient in most locations tested, as harmonics of the power line dominated the acquired signals. Additional shielding was provided inexpensively by placing the coil assembly inside a 30 cm length of 15 cm diameter, 7 mm wall aluminum tubing (Metal Supermarkets, Vaughan, ON). A thick shield is required as the skin depth at these frequencies is substantial, approximately 2 mm in aluminum at 2 kHz. With this shielding in place, the NMR signal dominates the spectrum, as can be seen in figure 4(b). Adding end caps to the shield tube made little further improvement.

Before the shielding was installed, an attempt was made to eliminate the power-line pickup by synchronizing the acquisitions with the power line. A low-voltage ac transformer powered from the mains was connected to one of the Arduino's digital inputs, and the pulse sequence triggered from the power line. With the phase cycling used, pickup from the power line is then subtracted on alternate transients. While this technique was found to work very well for short times, phase variations in the power line on time scales of ~ 0.5 s prevented effective cancellation for long-lasting signals.

3. Results and discussion

NMR signals acquired with the device described above, from a 0.55 L sample of tap water, are shown in figure 4. The free-induction decay in figure 4(a) is the result of averaging four

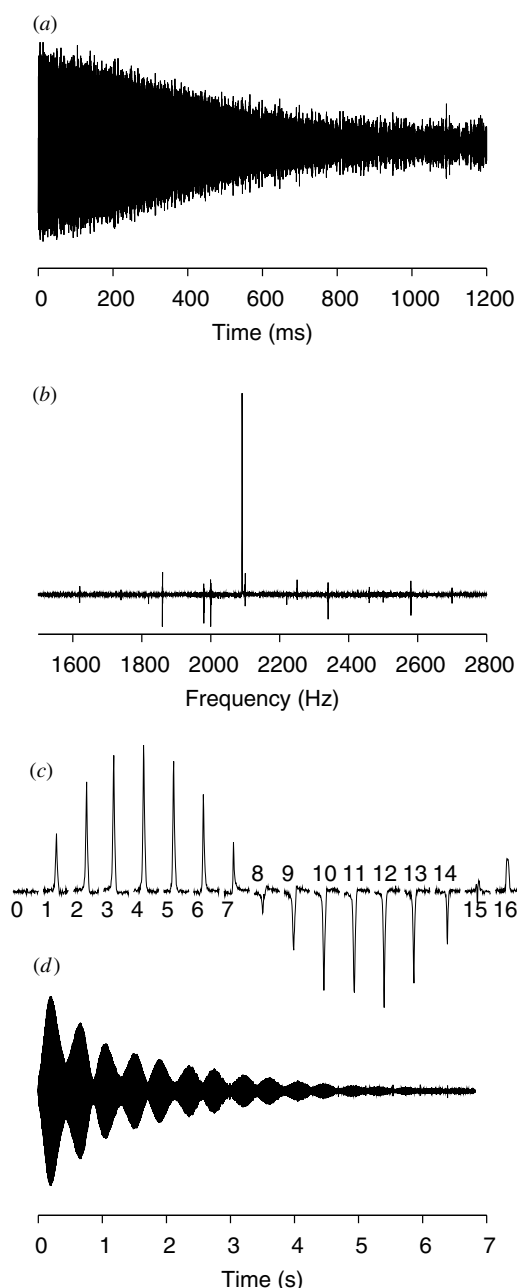


Figure 4. NMR signals. (a) The decay following a spin-echo pulse sequence. (b) The spectrum corresponding to (a), and (c) the variation in the signal observed as the duration of the initial pulse is varied. The numerical labels indicate the number of full audio-frequency cycles applied. (d) The result of a Carr-Purcell-Meiboom-Gill multi-echo sequence with 16 echoes. (a) and (b) are the results of four transients, each spectrum in (c) was acquired with two transients, while (d) represents 256 transients. Most of the noise spikes in (b) correspond to odd harmonics of the 60 Hz power-line frequency.

transients of a spin-echo pulse sequence, with a 10 ms echo time. The spectrum corresponding to this time-domain data is shown in figure 4(b). The data were zero filled from 16 384 to 32 768 points, Fourier transformed and phase adjusted, but not otherwise manipulated. A nutation plot is shown in figure 4(c). The phase adjustment applied to all spectra in figures 4(b) and (c) was identical. The small variation in phase between the

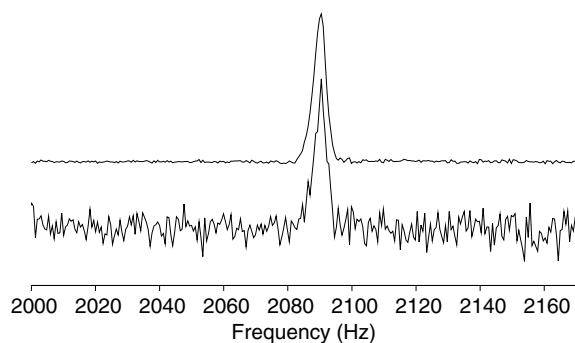


Figure 5. Polarization enhancement from the polarization coil. The lower trace is the result of 7500 transients with a 7 s recycle delay. The upper trace is the result of 44 transients with the polarization current on for 7 s. The vertical and horizontal scales of both data sets are identical, but the traces are offset vertically for clarity.

various spectra in the nutation plot is due to a small (~ 10 Hz) frequency offset from resonance of the transmitter. In each case, a simple phase cycle in which the phase of the first pulse was alternated between zero and π , and signals alternately added and subtracted was used. The phase of the inversion pulse was fixed.

The multiple pulse capability of the system is demonstrated in figure 4(d), where the field homogeneity was purposefully degraded slightly and a Carr–Purcell–Meiboom–Gill sequence [12] was used to create a train of 16 echoes. Signal intensity remains visible 7 s after the initial 90° pulse.

The enhancement of the signal achieved by polarizing in the polarization coil was measured by comparing the signals observed with the polarizing coil active to those without. These data are shown in figure 5. Integrals of the NMR peaks in these data indicate an enhancement of approximately 182, compared to the 212-fold increase in the polarization field, indicating that relatively little magnetization is lost in the adiabatic field turnoff. This also demonstrates the stability of the device, as the spectrum acquired without the polarization coil averaged over 14 h.

Callaghan *et al* have discussed the calculation of the SNR expected from their device in detail [10], and the calculation for the present device is similar. The details are discussed in the appendix. The magnitude of the signal observed is in excellent agreement with that expected, while the noise measured in an indoor urban environment is some 2.2 times larger than calculated. We find a typical time-domain SNR of 24 for a single-shot experiment in an indoor urban environment, with much of this noise arising from isolated power-line harmonics. The most important difference between Callaghan’s device and the present device in terms of sensitivity is that the polarization field used here is about a factor of 3 less strong, and additional shielding is used here. As can be seen by comparing figure 4(a) to Callaghan’s figure 7(b), which shows the average of four times as many transients with a polarization field three times larger, the present device compares very well. Despite the sixfold disadvantage from the reduced polarization field and smaller number of transients, the SNR here appears to be similar or better. This difference, however, is entirely due to the

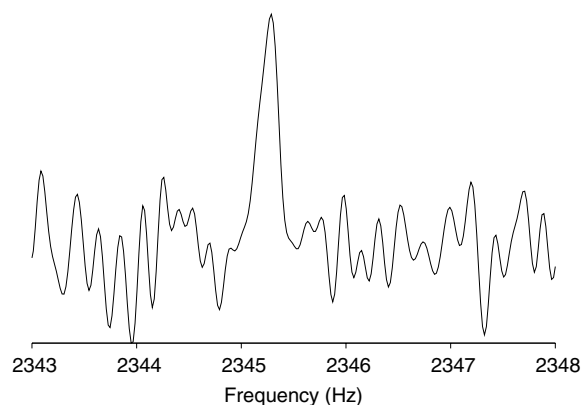


Figure 6. Earth’s field NMR spectrum of thermally polarized water.

increased shielding provided by the aluminum tubing. In both cases the receiver contributes negligibly to the noise.

The SNR observed in the frequency domain depends sensitively on the homogeneity of the magnetic field over the sample volume. The homogeneity shows an extreme dependence on the location of various objects nearby, and in fact can be improved by ‘shimming’ with objects at hand. The linewidth (full-width at half-maximum) in figure 5 is about 5 Hz, but in figure 4, an aerosol spray can (of compressed air) was moved to about 30 cm distant from the coil assembly, yielding a linewidth of ~ 0.6 Hz. In the outdoors, significantly narrower linewidths are more easily achieved. In figure 6, a spectrum that was acquired at a local beach is shown. This spectrum was acquired with thermal polarization in the Earth’s field, i.e. with no current in the polarization coil, and is the average of 64 transients, acquired over about 10 min. The full-width at half-maximum of the line is about 0.21 Hz, not much greater than the best possible linewidth of 0.18 Hz, based on the measured T_2 of 1.75 s.

Spectra and T_1 relaxation measurements have also been collected of an apple (145 g), a banana (128 g) and an intact aluminum can containing 355 mL of beer. The T_1 measurements were accomplished by adding a variable delay following the polarization period before applying the audio-frequency pulse and fitting the resulting magnetization decays to decaying exponentials. With substantial variation in signal intensity and minor variations in lineshape, attributable to the different geometries of the samples, the spectra are essentially indistinguishable from that of the water sample. In contrast to the spectra, the T_1 times of these samples vary significantly. For the tap water sample $T_1 = 2.3 \pm 0.1$ s, while for the beer it is notably shorter, 1.5 ± 0.15 s. For the apple, $T_1 = 1.0 \pm 0.1$ s, and the banana has $T_1 = 690 \pm 55$ ms. While the choice of these samples may appear whimsical, a great deal of research has been undertaken in food science with NMR (see [13] for example), and many more research opportunities exist in this area. A widely available low-cost spectrometer may accelerate such work. The fact that NMR measurements are possible on a liquid in an enclosed aluminum container is a testament to the ability of these low-frequency electromagnetic signals to penetrate conductors that would completely shield the radio frequencies used in high-field NMR.

The device used to collect all of the data shown above was built in a form most suitable for a teaching lab, as that is its intended use. The transmitter and receiver were constructed on a single $5 \times 16 \text{ cm}^2$ breadboard, while the polarization coil driver, due to the high current handling required of its components, was soldered together on a prototyping board. With a little effort at packaging the device in a robust manner, it could easily be configured for regular use in the field. It is small enough to be transportable in a single trip by bicycle. The transmitter/receiver combination draws a total of 16 mA from its supplies, making it suitable for battery powered operation, two 9 V batteries would suffice for ~ 20 h. The Arduino board is powered by its USB connection to a notebook computer. The computer power supply for polarizing the nuclei however requires on the order of 100 W. This requirement is easily met with a modest lead acid battery and inverter. Alternatively, the polarization coil could be powered directly from such a battery, either with the coil connections in parallel for a 6 V battery, or in series for a 12 V battery. A 6 V 12 A h sealed lead acid battery has a mass of less than 2 kg (e.g. Panasonic LC-R0612P, available from Digikey), and two of these in parallel could power the polarization coil at 50% duty cycle for more than 2 h.

As described, the present device is capable of producing any desired sequence of phase-coherent audio-frequency pulses, and so can be used for performing a wide variety of pulsed NMR experiments, including one-dimensional spectroscopy, relaxation measurements such as inversion recovery, spin-echo, and Carr-Purcell-Meiboom-Gill sequences, as well as two-dimensional heteronuclear (e.g. ^{19}F - ^1H) NMR experiments. Potential field applications include measuring moisture content and nuclear-spin relaxation parameters (of, for example, soil, fruit, ice, etc) and geomagnetic studies of the spatial and/or temporal variation of the Earth's field. In the teaching lab, the device could be provided fully assembled for use in exploring NMR and spin physics, or could be presented only partially complete, where students would build some significant fraction of the circuitry and the focus would be more on instrument building. At present the device has no pulsed-field gradient capability. Adding one field gradient for diffusion measurements or multiple gradients for imaging would be relatively straightforward. The Arduino has a number of additional digital outputs to facilitate control of these devices.

4. Conclusions

In conclusion, we have presented a low-cost, easy-to-build Earth's field NMR spectrometer suitable for use in research and teaching environments. The only specialized piece of equipment used in the construction was a coil-winding machine that was used to wind the 3100-turn detection coil, although a resourceful builder without access to such a machine could surely wind an acceptable coil by hand or with an improvised coil winder. A key component of this device is an inexpensive, easy-to-program microcontroller board that acts as a high-performance pulse programmer, audio-frequency synthesizer and ADC. The low cost, accessibility

of components and simplicity make such a device available to any researcher or student.

Acknowledgments

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Appendix. Signal and noise calculations

Ideally, the noise in an inductively detected NMR experiment is dominated by Johnson thermal noise from the coil resistance, and is given by [10]

$$V_{\text{Johnson}} = \sqrt{4k_B T \Delta f R} \quad (\text{A.1})$$

where k_B is Boltzmann's constant, T is the temperature of the coil, Δf is the noise bandwidth and R is the coil resistance.

The signal expected can be expressed as

$$V_{\text{signal}} = \left(\frac{B_1}{i_1} \right) V \omega_0 M_0 \quad (\text{A.2})$$

where B_1/i_1 is the field produced per unit current in the detection coil, V is the sample volume, ω_0 is the Larmor frequency and M_0 is the sample magnetization, given by

$$M_0 = \frac{N_s \gamma^2 \hbar^2 B_p}{4k_B T}. \quad (\text{A.3})$$

Here, N_s is the number of protons per unit volume of sample, γ is the gyromagnetic ratio of the proton and B_p is the polarization field.

Under ideal conditions, the SNR is given simply by the ratio of equation (A.2) to equation (A.1). For a sample of fixed dimensions and coil wound from copper wire, the optimal SNR is set by the total mass of the conductor used in the receive coil. This is made clear by comparing a coil made with a single layer of wire to a two-layer coil wound with wire of one-half of the diameter. The two-layer coil will have four times greater B_1/i_1 (twice as many turns per layer and two layers), and so the received signal is a factor of 4 greater. On the other hand, the resistance per unit length of the wire is a factor of 4 greater, and the total length of wire needed to wind the two layers is a factor of 4 greater. The result is that the noise in the two layer coil is also increased by a factor of 4, and the SNR is unchanged.

While winding a single-layer coil is much easier than a multi-layer coil, there are compelling reasons for using more layers. The larger voltages for both signal and noise from the multi-layer coil enable the coil thermal noise to completely dominate the receiver amplifier noise. Also, the higher inductance reduces the tuning capacitance needed. A practical upper limit to the number of turns used is set by the self-resonance frequency of the coil.

Important deviations from the ideal behavior come from two sources. The first is from pickup of environmental noise, which in an 'urban' environment will often overwhelm the Johnson noise of equation (A.1) by several orders of

magnitude. The second is due to changes in the coil performance due to coupling of the receive coil with other inductive circuits or conductors, such as the polarization coil or shielding, nearby.

The shielding of the receive coil from environmental electromagnetic noise is in principle straightforward. Placing the coil within a conducting enclosure will exclude the environmental noise. The construction of the shield however demands a number of design trade-offs. An effective shield must be several skin-depths thick at the relevant frequencies. The skin depth of aluminum is approximately 2 mm at 2 kHz. A shield attenuating pickup by 95% would need to be three skin depths, or 6 mm thick. The skin depth of copper at the same frequency is 1.5 mm, making a copper shield of similar effectiveness somewhat thinner, but because of copper's much higher density and cost, it would be much heavier and much more expensive. A much thinner shield could be constructed from mild steel, but the magnetic properties would unacceptably perturb the Earth's magnetic field. To minimize the size and weight of the shield, one would naturally make it as close as possible in size to the receive coil. Unfortunately, such a shield impacts the performance of the spectrometer as will be described below.

Here, the coupling of the detection coil with the polarization coil increases the effective resistance of the detection coil. In the coil configuration used, the response of the circuit, as measured by the frequency-dependent voltage induced by the field from a small test coil, indicates an effective resistance of 800 Ω . The Q , or quality factor, of the circuit measured this way is 5.8.

Inserting the filter bandwidth of 212 Hz, equation (A.1) suggests a root-mean-square (rms) noise of 53 nV. This noise is amplified by the circuit Q of 5.8 and the receiver gain of 50 000, and then digitized by a 10 bit digitizer with a 5 V full scale. We report signal and noise measurements from the instrument in terms of least significant bits (LSB) of the ADC, as those most directly reflect the data acquired from it. Each LSB corresponds to 5 V/1024 $\sim$ 4.88 mV. The above calculation yields an expected thermal noise corresponding to 3.2 LSB.

Quantization and nonlinearity imperfections of the ADC are estimated to contribute another 1.25 LSB, added in quadrature with the thermal noise, resulting in an expected rms noise of 3.4 LSB. Noise collected with the coil resonant circuit yields an rms noise of 7.5 LSB. Much of the 2.2-fold excess noise is concentrated in a small number of harmonics of the power-line frequency, eroding the SNR in the time domain. These harmonics however do not significantly affect observation of the signal in the frequency domain. Digitally removing the three largest power-line harmonics brings the time-domain noise down to 5.2 LSB, approximately 53% larger than the expected ideal. Given the indoor urban environment and modest shielding used, this is relatively good agreement.

Noise from the receiver and T/R switch relays is negligible, as is demonstrated by the reduction of the noise to an rms value of 1.4 LSB when the resonant detection circuit is replaced with a 2 k Ω resistor. In this case, the noise is

dominated by the quantization noise of the ADC. The reduction from 7.5 LSB to 1.4 LSB indicates a receiver noise figure of <0.08 dB; a noiseless amplifier would increase the SNR by no more than 2%.

An approximate numerical calculation for B_1/i_1 gives a value of 23 mT A⁻¹, in reasonable agreement with the 24.7 mT A⁻¹ value given in [10]. However, the coupling between the detection coil and polarization coil further perturbs the coil performance, and a direct measurement of B_1/i_1 found from the measured current (0.833 mA, peak to peak) and the nutation frequency (130 Hz) indicates $B_1/i_1 = 14.7$ mT A⁻¹. Taking a sample volume of 0.55 L, $B_p = 10.5$ mT, and discounting the polarization by 0.86 to account for losses in the adiabatic polarization field turnoff, as described earlier, we expect a signal amplitude of 3.06 μ V, or 182 LSB in one transient. The maximum signal observed following a single pulse, 179 LSB, is in excellent agreement with the calculation.

To summarize the effects of the coupling between the detection and polarization coils: the noise is increased, as indicated by the fact that the detection coil effective resistance (800 Ω) is greater than the DC resistance (310 Ω) and the received signal is decreased due to the reduction in B_1/i_1 . The net effect of these two factors is a reduction in the SNR by a factor of 2.5 compared to that expected from an uncoupled coil. In return, however, the shielding provided by the polarization coil provides orders of magnitude reduction in pickup of environmental noise. This factor of 2.5 was expected, as it is identical to the factor described in [10], and the detection coil design here closely follows the design there.

Beyond the factor of 2.5, the time-domain SNR here is reduced by a factor of 2.2 due to the pickup of environmental noise. As described above, most of this noise is concentrated in a small number of power-line harmonics. When the three most significant harmonics are removed, the time-domain SNR reduction due to environmental noise becomes a factor of 1.53. In comparison, Callaghan reports the same factor of 2.5 due to coil coupling, and an additional factor of 1.36 due to environmental noise pickup, but those measurements were made in one of the most electromagnetically quiet environments on the planet, Antarctica, while the measurements here were made in a busy research building located in a major city. Callaghan's SNR calculation predicts an SNR a factor of $\sim$ 3 greater than that here due to the threefold greater polarization field used.

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